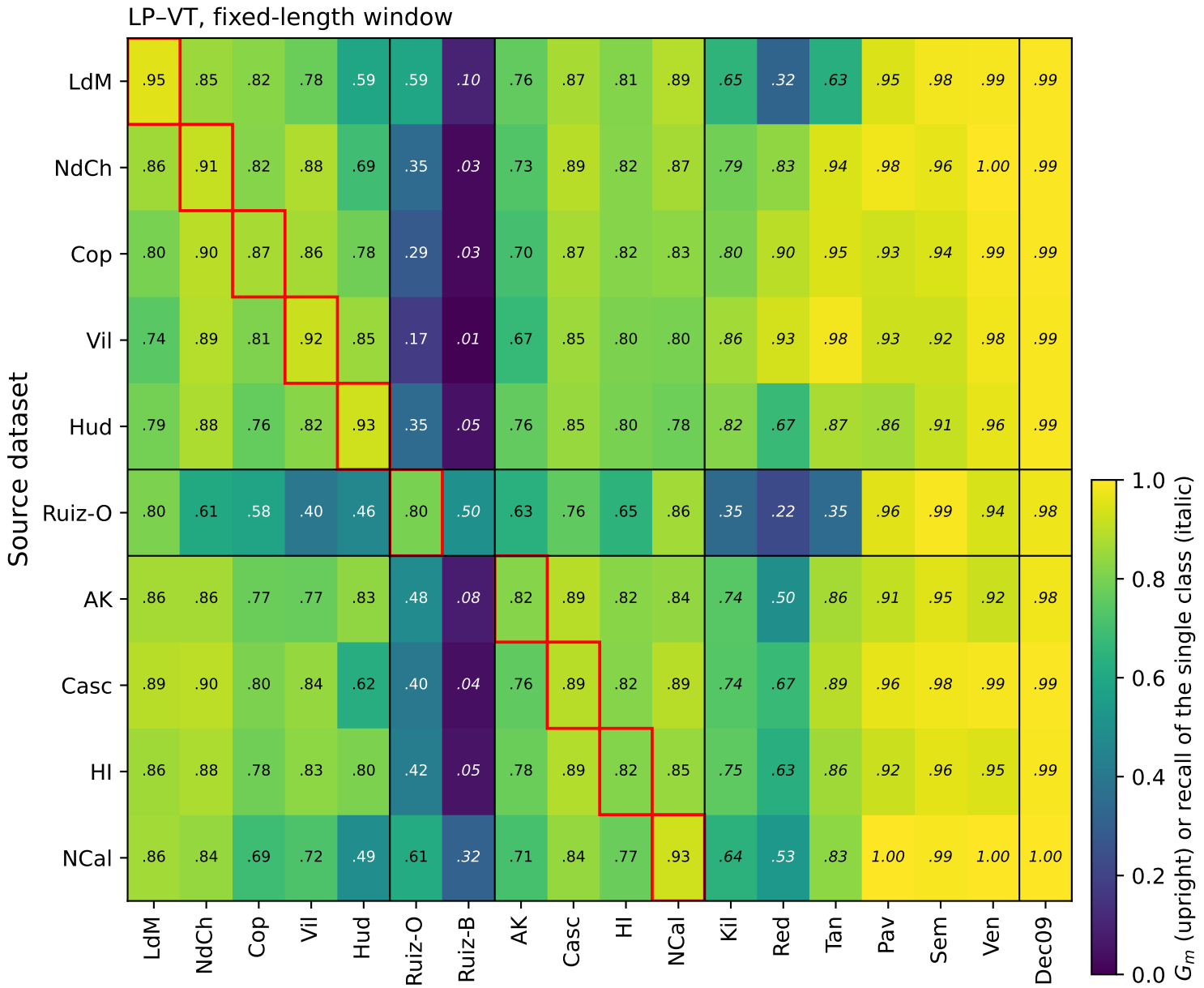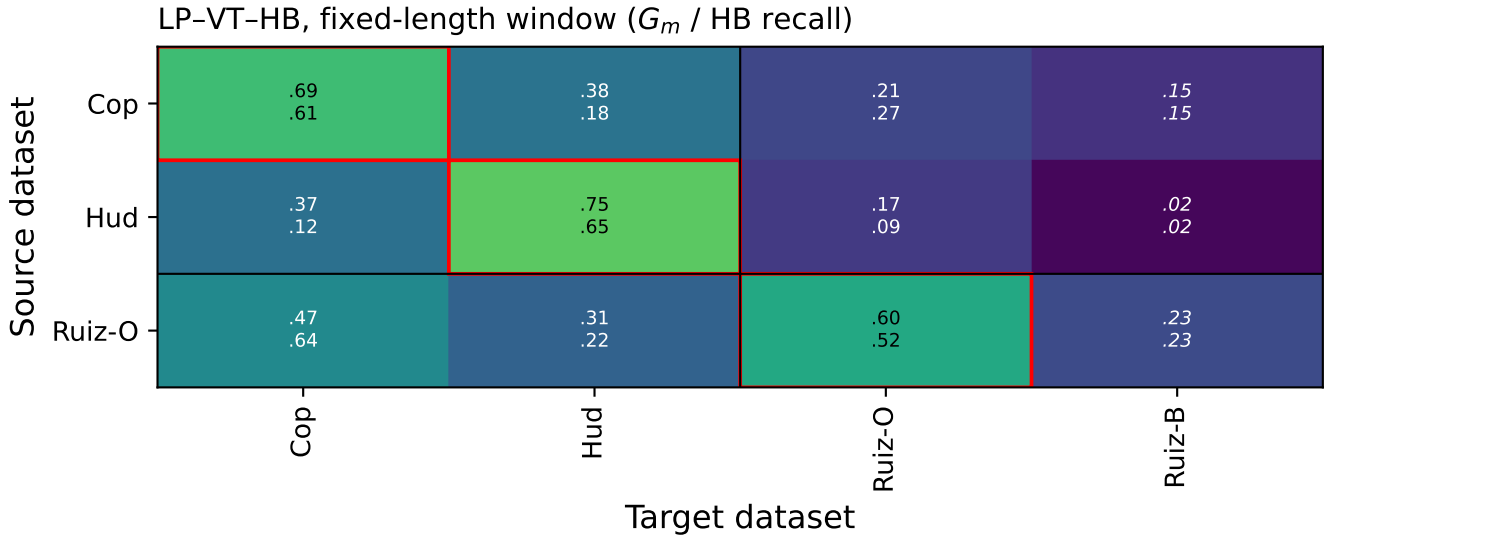Transfer matrices, gradient-boosted trees; red: within-dataset validation